

Geometry-Driven Triangulation and Differentiable Semantic Gaussian Refinement for 3D Grape Bunch Model Reconstruction in the Field

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Abstract

Berry thinning in table-grape production requires estimating berry count and bunch structure under severe occlusion. While existing 3D reconstruction and neural rendering pipelines can recover dense geometry or enable novel view synthesis, they rarely produce compact, instance-level models for densely packed grape bunches. We propose a geometry-aware, three-stage framework that reconstructs a semantic 3D model of individual berries from handheld field video. Each berry is represented as an opaque 3D Gaussian ellipsoid with a 3D center, principal radii, and orientation. Stage 1 applies a berry detection/segmentation network together with a lightweight structure-from-motion (SfM) COLMAP method run to obtain multi-view 2D attributes (centers, circle/ellipse parameters, masks, and visibility labels) and initial camera poses; these 2D observations later act as supervision for optimizing the 3D model. Stage 2 forms multi-view tracks using epipolar constraints, azimuth-based neighborhood restriction, and radius consistency (without appearance descriptors), then triangulates and consolidates 3D berry centers via geometric gating and 3D non-maximum suppression. Stage 3 refines the compact semantic model by differentiable rendering under multi-head 2D losses, with conservative clone-prune topology updates driven by persistent unmatched evidence. On 30 real-field bunches with labelled point-cloud ground truth converted to 3D ellipsoids, the refined models achieve F1 scores of 78.7–88.5% at oriented bounding box Intersection of Union (OBB-IoU) threshold (τ) at 0.10 and 64.5–74.9% at $\tau = 0.25$, while Stage 3 runs in 1.8–5.1 minutes per bunch for 10–30 selected views. These results

show that combining geometry-driven initialization with differentiable, instance-level refinement yields robust and interpretable per-berry 3D reconstructions from unconstrained vineyard video.

CCS Concepts

• Computing methodologies; • Computer vision; • Rendering;

Keywords

3D semantic reconstruction, precision viticulture, multi-view geometry, differentiable rendering, 3D Gaussian primitives

ACM Reference Format:

Yan San Woo, Kabin Yok-In, Prawit Buayai, Hiromitsu Nishizaki, Latifah Munirah Kamarudin, and Xiaoyang Mao. 2026. Geometry-Driven Triangulation and Differentiable Semantic Gaussian Refinement for 3D Grape Bunch Model Reconstruction in the Field. In *2026 The 9th International Conference on Information and Computer Technologies (ICICT 2026)*, March 11–13, 2026, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3803291.3803356>

1 INTRODUCTION

Table-grape quality depends strongly on bunch compactness and berry size, both of which are regulated during berry thinning. In practice, workers must estimate the number and spatial arrangement of berries under heavy occlusion and strong self-similarity, which makes thinning cognitively demanding and leads to variability between operators.

Computer-vision tools have been proposed for berry detection and 2D counting from single images [1–4], but their outputs are inherently viewpoint-dependent and do not provide the 3D structure needed for robust phenotyping or downstream automation. At the other extreme, general 3D pipelines such as classical structure-from-motion / multi-view stereo (SfM/MVS) and neural rendering (e.g. NeRF [5] and 3D Gaussian Splatting [6]) recover dense scene geometry or radiance fields, yet typically lack a compact, instance-level

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ACM ISBN 979-8-4007-2252-3/2026/03

<https://doi.org/10.1145/3803291.3803356>

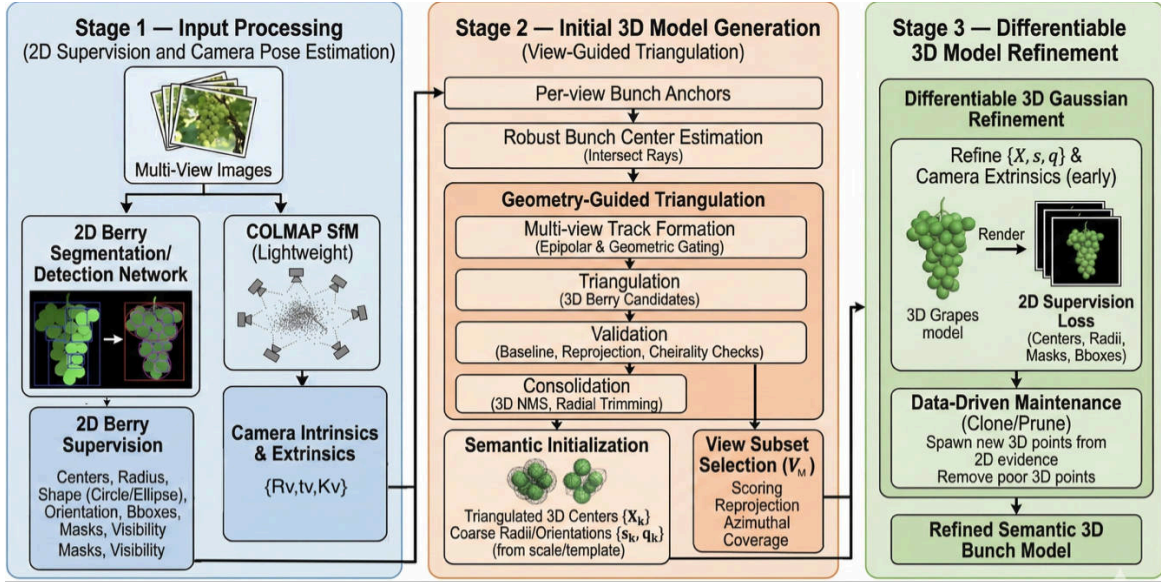


Figure 1: Proposed three-stage pipeline. Stage 1: multi-view 2D supervision and initial camera poses from detection/segmentation and lightweight SfM. Stage 2: geometry-aware matching and triangulation with 3D NMS to initialize berry centers. Stage 3: differentiable refinement of 3D Gaussian berries with joint pose warm-up and clone-prune topology maintenance.

representation and often require controlled capture. Extracting per-berry centers, sizes, and identities from such dense representations remains challenging in cluttered vineyard scenes.

This work targets instance-level 3D semantic reconstruction of grape bunches from short handheld multi-view sequences in real vineyards. Each berry is modeled as a compact parametric primitive (3D center, radii, orientation). We combine geometry-aware triangulation from multi-view 2D detections with differentiable refinement of a small set of 3D Gaussian berries guided by multiple 2D cues (centres, masks, boxes, and shape/scale).

Contributions. We introduce

1. a geometry-aware triangulation pipeline that builds multi-view berry tracks from 2D detections and lightweight SfM method COLMAP, then initializes per-berry 3D centers via robust triangulation and 3D non-maximum suppression
2. differentiable refinement of a compact set of 3D Gaussian berries using multi-head 2D losses and a warm-up phase that refines camera poses; and
3. a conservative clone-prune strategy that adds berries supported by persistent unmatched evidence and removes consistently unsupported hypotheses, improves berry-count accuracy and 3D oriented bounding-box Intersection of Union (OBB-IoU) on 30 real-field bunches at practical runtimes.

An overview is shown in Figure 1.

2 RELATED WORK

3D reconstruction for agriculture has used classical SfM/MVS pipelines to recover geometry from multi-view images, but performance in vineyards is often limited by foliage clutter, repeated textures, motion blur, and occlusion, producing noisy or incomplete point clouds that require additional segmentation or clustering for

object-level analysis. Neural rendering methods such as NeRF [5], pose-refining extensions [7], and 3D Gaussian Splatting [6] optimize dense scene representations from posed images and excel at view synthesis and fine detail, but their outputs are dense and entangled; obtaining compact, instance-level semantics typically requires external masks or post-hoc clustering, which is difficult when many small, similar berries overlap in depth.

For grape bunches specifically, prior work includes laboratory systems based on laser scanning, stereo, or structured light that yield accurate point clouds under controlled backgrounds but are not field-deployable [8, 9], and image-based methods that exploit spherical priors or simplified geometry to estimate 3D structure from one or a few views [2, 10]. More recent pipelines combine SfM with berry segmentation and point-cloud clustering (e.g., K-means) to initialize per-berry hypotheses and refine them via reprojection losses [11], but they are sensitive to pose error and point-cloud quality and depend on external berry-count estimates to set the number of clusters.

3 PROPOSED METHOD

Given a handheld multi-view sequence $\{I_v\}_{v=1}^V$ of a single grape bunch, our goal is to reconstruct a compact semantic 3D model

$$\mathcal{B} = \{(X_k, s_k, q_k)\}_{k=1}^K$$

where $X_k \in \mathbb{R}^3$ is the 3D center of berry k , $s_k = (s_{x,k}, s_{y,k}, s_{z,k})$ are principal-axis Gaussian radii, and $q_k = (q_{x,k}, q_{y,k}, q_{z,k}, q_{w,k})$ is a unit quaternion encoding orientation. Each berry is therefore modeled as an opaque 3D Gaussian ellipsoid, and the set $\mathcal{B}$ forms the semantic 3D bunch model.

The pipeline has three stages as illustrate in Figure 1

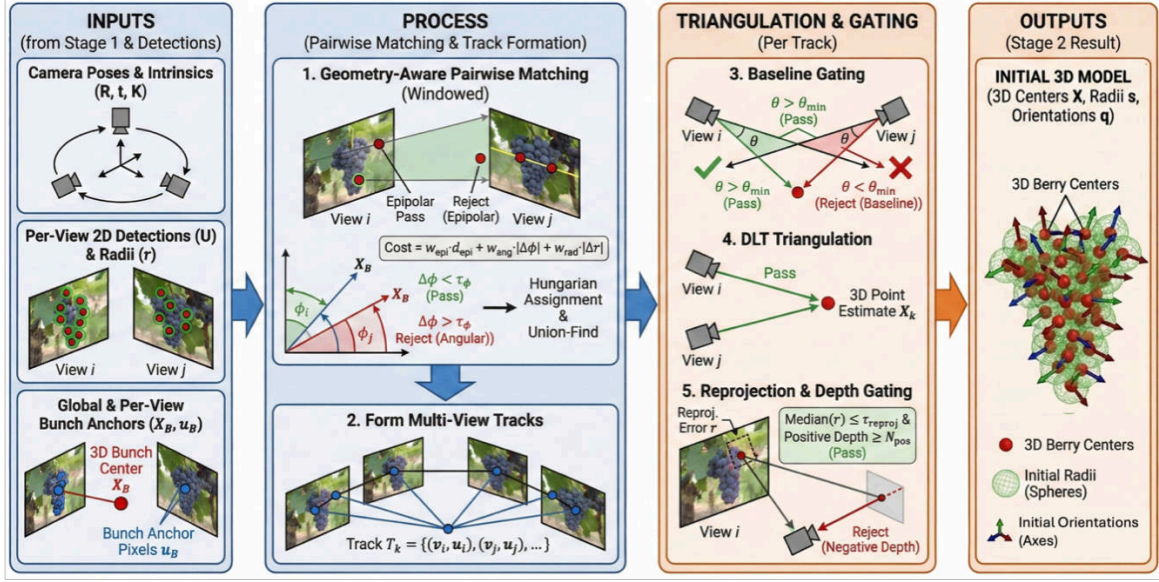


Figure 2: Stage 2: Initial 3D model generation (view-guided triangulation flow).

Stage 1 – input processing: extract multi-view 2D berry observations (centres, shape/scale, masks, boxes, visibility labels) and initial camera parameters (intrinsics and extrinsics) from a lightweight SfM method COLMAP run.

Stage 2 – initial 3D model generation: build geometry-aware multi-view tracks from the 2D detections, triangulate 3D berry centres under several gating checks, and consolidate nearby hypotheses into a compact set of 3D seeds.

Stage 3 – differentiable refinement: refine the semantic Gaussian berries via differentiable rendering with multi-head 2D losses, a short pose-refinement warm-up, and a conservative clone-prune mechanism for topology maintenance.

4 Stage 1: Input Processing (2D Supervision and Camera Pose Estimation)

Each image I_v is processed by the berry detection/segmentation network of [4], yielding for each detected berry i in view v a bounding box $B_i^{(v)}$ and an instance mask $M_i^{(v)}$. From these outputs we derive per-berry 2D attributes that will later be used to supervise Stage 3, the 3D optimization. For each detection we extract: (i) center $u_i^{(v)} = (u_i^{(v)}, v_i^{(v)})$; (ii) shape/scale, represented either as a circle radius $r_i^{(v)}$ or ellipse semi-axes $(r_{x,i}^{(v)}, r_{y,i}^{(v)})$; (iii) in-plane ellipse orientation $\alpha_i^{(v)}$, estimated by ellipse fitting to $M_i^{(v)}$ when an elongated footprint is detected; and (iv) a visibility label (visible / partially occluded / not reliably visible) used to gate losses. The bounding boxes $B_i^{(v)}$ and masks $M_i^{(v)}$ are also used for box- and mask-level losses in Stage 3.

In parallel, we estimate camera parameter using a lightweight SfM method COLMAP run configured with a single OPENCV camera model. COLMAP reports one set of intrinsics K_v (focal lengths, principal point, and distortion coefficients) shared by all views, and

per-view extrinsics (R_v, t_v) that describe the camera pose in the world coordinate system (rotation and translation from world to camera coordinates). To reduce computation relative to a full SfM method COLMAP configuration, global bundle adjustment is disabled. All geometry is expressed in this COLMAP world coordinate system so that triangulation (Stage 2) and differentiable rendering (Stage 3) are consistent. These poses are treated as an approximate initialization: Stage 2 uses them for geometric constraints, and Stage 3 refines extrinsics (R_v, t_v) during a warm-up phase.

Stage-1 output therefore consists of:

- per-view 2D supervision

$$\{B_i^{(v)}, M_i^{(v)}, u_i^{(v)}, r_i^{(v)} \text{ or } (r_{x,i}^{(v)}, r_{y,i}^{(v)}), \alpha_i^{(v)}, \text{vis}_i^{(v)}\}; \text{ and}$$

- camera parameters $\{K_v, R_v, t_v\}_{v=1}^V$ for all views v .

4.1 Stage 2: Initial 3D Model Generation (View-Guided Triangulation)

Stage 2 converts per-view detections into an initial semantic seed model as shown in Figure 2. Because berries are visually repetitive and weakly textured, we avoid appearance descriptors and rely purely on geometry.

4.1.1 View Filtering and Robust Bunch Center. Let N_v be the number of detected berries in view v . We define $N_{\max} = \max_v N_v$ and $N_{\min} = \tau_{\text{det}} N_{\max}$; views with $N_v < N_{\min}$ are discarded. We use a scalar threshold $\tau_{\text{det}} = 0.6$ in all experiments, which removes frames with very weak detections while retaining most informative views.

For each remaining view we define a bunch anchor $u_B^{(v)}$ as the median of its berry centers. We back-project this pixel into a world-space ray using the COLMAP camera:

$$\tilde{d}_v^{\text{cam}} = K_v^{-1} \tilde{u}_B^{(v)}, d_v = \frac{R_v^T \tilde{d}_v^{\text{cam}}}{\|R_v^T \tilde{d}_v^{\text{cam}}\|_2},$$

where K_v is the intrinsic matrix of view v , R_v is its rotation, $\tilde{u}$ denotes homogeneous coordinates and d_v is a unit ray direction in world coordinates. Let C_v be the camera center in world coordinates. The 3D bunch center X_B is estimated as the point with minimum total orthogonal distance to all rays:

$$X_B = \arg \min_X \sum_v \|(I - d_v d_v^T)(X - C_v)\|_2^2, \quad (1)$$

We then compute each view's azimuth ϕ_v around X_B . These azimuths are used for neighborhood restriction and angular gating in the matching stage.

4.1.2 Geometry-Aware Pairwise Matching and Multi-View Tracks. We construct correspondences between detections using four geometric cues.

- Neighborhood restriction (azimuth window)

Views are sorted by azimuth ϕ_v . Each view i is matched only to views whose indices differ by at most 4 (i.e., $|\Delta \text{index}| \leq 4$). This preserves parallax while avoiding matching all possible view pairs.

- Epipolar gating

For a detection $\tilde{u}_{i\ell}$ in view i , candidate matches in view j must lie near its epipolar line

$$\ell_j = F_{ij} \tilde{u}_{i\ell},$$

where F_{ij} is the fundamental matrix from (K_i, R_i, t_i) and (K_j, R_j, t_j) .

The point-to-line distance

$$d_{\text{epi}}(\tilde{u}_{i\ell}, \tilde{u}_{jm}) = \frac{|\tilde{u}_{jm}^T F_{ij} \tilde{u}_{i\ell}|}{\sqrt{a^2 + b^2}}, \ell_j = (a, b, c)^T$$

must satisfy $d_{\text{epi}} \leq \tau_{\text{epi}}$ with $\tau_{\text{epi}} = 3\text{px}$.

- Angular consistency

Candidate view pairs must satisfy

$$|\Delta\phi| = |\phi_i - \phi_j| < \tau_\phi,$$

with $\tau_\phi = 25^\circ$. This enforces a smooth change of apparent ordering around the bunch and excludes pairs that are too far apart on the view ring.

- Scale consistency

We softly prefer matches with similar normalized 2D radii $\hat{r}_i, \hat{r}_j$ (either circles or ellipse sizes, normalized to account for per-view scale differences).

These cues define a gated matching cost for a detection pair $(i, \ell) \leftrightarrow (j, m)$:

$$C_{ij}(\ell, m) = w_{\text{epi}} d_{\text{epi}} + w_{\text{ang}} |\Delta\phi| + w_{\text{rad}} |\hat{r}_{i\ell} - \hat{r}_{jm}|,$$

with weights $w_{\text{epi}} = 1.0$, $w_{\text{ang}} = 0.1$, $w_{\text{rad}} = 0.02$.

Pairs that violate epipolar or angular thresholds are forced to a large constant cost $C_{\text{max}} = 10^6$.

For each view pair (i, j) in the azimuth window, we solve a Hungarian assignment on the cost matrix C_{ij} to obtain an optimal one-to-one matching between detections (Hungarian = standard minimum-cost bipartite matching algorithm). Accepted matches across all view pairs are then aggregated using a union-find (disjoint-set) structure, which merges all detections connected by matches into multi-view tracks

$$\mathcal{T}_n = \{(v_k, \ell_k)\}_{k=1}^{M_n}.$$

Tracks observed in fewer than three views ($M_n < 3$) are discarded.

4.1.3 Triangulation with Geometric Validation and 3D Consolidation. For each track $\mathcal{T}_n$ we triangulate a 3D berry-center hypothesis X_n .

Let $\{P_{v_k}\}$ be the projection matrices

$$P_{v_k} = K_{v_k} [R_{v_k} \ t_{v_k}]$$

and $\{u_k\}$ the corresponding image points. We form a standard Direct Linear Transform (DLT) system $AX = 0$ from the cross-product constraints $u_k \times (P_{v_k} X) = 0$ and solve for a homogeneous point $\tilde{X}_n$; and obtain the Euclidean 3D point

$$X_n = \frac{\tilde{X}_n}{\tilde{X}_{n,4}}.$$

We then apply three geometric checks:

- Minimum baseline angle

Using the world-space ray directions $\{d_k\}$ associated with $\mathcal{T}_n$, compute

$$\theta_{\min}(X_n) = \min_{a \neq b} \angle(d_a, d_b).$$

Tracks with $\theta_{\min}(X_n) < \Theta_{\min}$ are rejected as ill-conditioned; we use a baseline threshold $\Theta_{\min} = 1.5^\circ$.

- Reprojection error

We project into each view,

$$\hat{u}_k = \pi(P_{v_k} X_n), r_k = \|u_k - \hat{u}_k\|_2,$$

and require the median reprojection error $\tilde{r}(X_n) = \text{median}_k(r_k) \leq \tau_{\text{reproj}}$, with threshold $\tau_{\text{reproj}} = 3\text{px}$.

- Cheirality

Depth must be positive in at least M^+ views:

$$z_k = (P_{v_k} X_n)_z > 0, \# \{k : z_k > 0\} \geq M^+.$$

where $(\cdot)_z$ denotes the third (depth) component.

Triangulation often yields duplicates when several tracks correspond to the same physical berry. We therefore apply a simple 3D non-maximum suppression (NMS) in Euclidean space. Let $\{X_n\}_{n=1}^N$ be the set of triangulated points that pass the above gates. Using a consolidation radius $\rho_{\text{nms}} \approx 8 - 10 \text{ mm}$, we are greedily:

1. Pick an unassigned point as a new representative.
2. Group all points within ρ_{nms} of this representative.
3. Replace the group by its centroid.

This yields a compact set of 3D berry centers $\{X_k\}$.

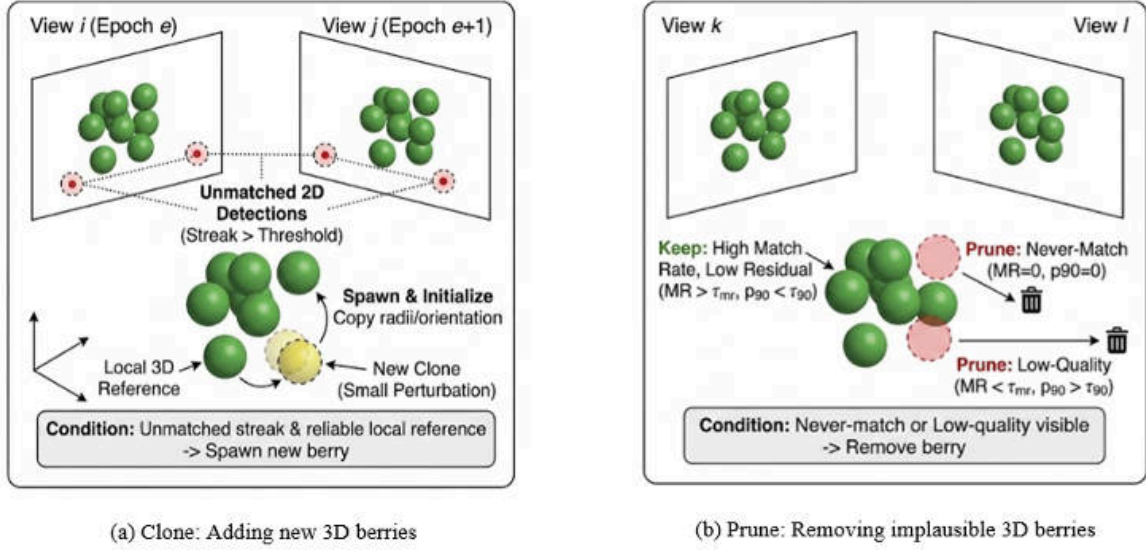


Figure 3: Data-driven topology maintenance (Clone and Prune).

4.1.4 *Initialization of Semantic Parameters and View Subset Selection.* For each 3D center X_k , we initialize the semantic parameters:

- Gaussian radii s_k from multi-view scale cues when stable (consistent 2D radii across views); otherwise from a dataset-derived spherical prior.
- Orientation q_k as the identity quaternion; q_k is kept fixed during refinement in our current system to reduce degeneracies between pose, scale, and orientation under 2D supervision.

Stage 3 does not need all remaining views. To trade off runtime and angular coverage we select a subset V_M with $M \in \{10, 20, 30\}$ by farthest-point sampling (FPS) over azimuths $\{\phi_v\}$, i.e., a greedy procedure that repeatedly chooses the view whose azimuth is furthest from those already selected.

The output of Stage 2 is therefore:

- The initial semantic seeds $\{(X_k, s_k, q_k)\}$, and
- The selected view subset V_M used for Stage-3 refinement.

4.2 Stage 3: Differentiable Refinement of Semantic Gaussians

Stage 3 takes the initial 3D semantic seeds from Stage 2 and refines it by differentiable rendering under multi-view 2D supervision. In parallel, a conservative clone-prune mechanism can add or remove berries when the 2D evidence is consistently under- or over-explained as shown in Figure 3.

4.2.1 Semantic Representation and Learnable Camera Extrinsic.

Each berry k is represented as an opaque 3D Gaussian ellipsoid parameterized by (X_k, s_k, q_k) , where $X_k \in \mathbb{R}^3$ is the center, $s_k \in \mathbb{R}^3$ are principal-axis radii, and $q_k \in \mathbb{R}^4$ is a unit quaternion encoding orientation. We initialize all berries as spherical Gaussians with $s_{x,k} = s_{y,k} = s_{z,k}$; anisotropic radii are only enabled later in training.

For each selected view $v \in V_M$, the camera intrinsics matrix K_v is fixed. The camera extrinsics (R_v, t_v) are initialized from COLMAP

and optimized only during an initial warm-up phase; afterwards they are then frozen so that topology updates (clone/prune) occur under a stable camera configuration.

The inputs to Stage 3 are thus the Stage-2 seeds $\{(X_k, s_k, q_k)\}$, camera parameters $\{K_v, R_v, t_v\}_{v \in V_M}$ restricted to V_M , and the 2D supervision from Stage 1 (centers, radii/ellipse parameters, masks, and boxes).

4.2.2 *Differentiable Rendering, Per-View Matching and Loss.* A differentiable renderer projects all Gaussians into each view $v \in V_M$. We use a standard z-buffer to determine which berry is visible at each pixel, so occlusions are handled naturally. For each rendered berry in view v we extract: (i) projected center $u_k^{(v)}$; (ii) projected shape/scale (circle radius $\hat{r}_i^{(v)}$ or ellipse semi-axes $(\hat{r}_{x,i}^{(v)}, \hat{r}_{y,i}^{(v)})$); (iii) predicted mask $\hat{M}_k^{(v)}$ and bounding box $\hat{B}_k^{(v)}$; and (iv) a visibility state (visible / partially occluded / out-of-bounds).

Because the number of visible berries differs per view, we establish correspondences by solving a Hungarian assignment between rendered instances and 2D detections in that view, restricted to visible and partially occluded, in-frame projections. The matches define a multi-head loss:

$$\mathcal{L} = \lambda_{\text{ctr}} \mathcal{L}_{\text{ctr}} + \lambda_{\text{rad}} \mathcal{L}_{\text{rad}} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} + \lambda_{\text{box}} \mathcal{L}_{\text{box}}.$$

where:

- $\mathcal{L}_{\text{ctr}}$: robust (Huber) loss on 2D centers,
- $\mathcal{L}_{\text{rad}}$: robust loss on radii / ellipse axes,
- $\mathcal{L}_{\text{mask}}$: segmentation loss (Dice + focal BCE) on masks,
- $\mathcal{L}_{\text{box}}$: IoU-type loss on bounding boxes.

Each term is evaluated only when the berry is labelled as visible and partially occluded in that view and the corresponding annotation is available.

4.2.3 Optimization Schedule and Clone–Prune Topology Maintenance. To avoid instability at the start of training, we adopt a three-phase schedule:

Phase 1 – Warm-up (pose refinement)

Optimize berry centers X_k and camera extrinsics (R_v, t_v) ; keep radii $\{s_k\}$ and orientations $\{q_k\}$ fixed. Clone–prune is disabled. We exit warm-up once the average per-view match rate (fraction of detections matched by the Hungarian step) exceeds a threshold (e.g., ≥ 0.40).

Phase 2 – Stable topology with clone–prune.

Freeze camera extrinsics (R_v, t_v) and continue optimizing X_k (and optionally a subset of radii). After a short buffer (e.g., 5 epochs), we enable a conservative topology update:

- Clone: if a 2D detection stays unmatched for at least 3 consecutive epochs and lies near a well-supported 3D neighbor, we spawn a new berry by copying that neighbor’s radii and orientation and adding a small 3D perturbation. A minimum spacing constraint and a cap on new clones per epoch prevents duplicates.
- Prune: for each 3D berry we track (i) match-rate and (ii) high-percentile residuals over a sliding window of epochs. Berries that are never matched or that have persistently low match rate and large residuals are removed, again with a per-epoch cap.

Phase 3 – Scale refinement

Once the loss plateaus, we unfreeze radii $\{s_k\}$ and jointly optimize (X_k, s_k) with camera extrinsics fixed. Orientations $\{q_k\}$ remains fixed in our current implementation.

The final output of Stage 3 is an image-aligned semantic 3D reconstruction $\{(X_k, s_k, q_k)\}$ whose berry count and geometry are jointly supported by all selected views.

5 EXPERIMENTAL AND RESULTS

We evaluate the proposed three-stage framework on 30 real-field table-grape bunch sequences captured with a handheld smartphone while moving around each bunch to approximate a view ring. The videos are recorded at a resolution of 1080×1920 , and all frames are processed at this native resolution before Stage-1 detection/segmentation. The berry detector/segmentor of [4] is applied to each frame to obtain per-berry masks and bounding boxes, as described in Stage 1: Input Processing (2D Supervision and Camera Pose Estimation).

For each bunch, we first run Stage 2 once to obtain the initial semantic 3D model, denoted INIT. Stage 3 is then executed three times using view subsets of size $M \in \{10, 20, 30\}$. The corresponding refined models are denoted OPT-10, OPT-20, and OPT-30, respectively. Unless otherwise stated, quantitative results are reported for both INIT and the OPT variants.

Stage-3 refinement is implemented in PyTorch and run on a single NVIDIA RTX 4080 SUPER (16 GB) GPU. We use SGD with learning rate 10^{-3} for berry parameters (X_k, s_k) and a smaller learning rate 10^{-5} for camera extrinsics (R_v, t_v) during the warm-up phase. Berry orientations q_k are kept fixed throughout refinement to avoid pose–scale–orientation degeneracies under purely 2D supervision. Reported Stage-3 runtimes correspond to the wall-clock time from INIT to the converged OPT- M model for each bunch

(Stage-1 and Stage-2 costs are shared across all M and are therefore not included).

5.1 Ground Truth 3D Semantic

Ground-truth 3D annotations are provided as labelled point clouds, where each 3D point is assigned a berry ID. For evaluation, every berry-specific point cluster is converted into a Gaussian ellipsoid via PCA: the centroid defines the 3D center X_k^{gt} , the principal axes define the orientation q_k^{gt} , and axis lengths are derived from the PCA variances with a global scaling factor $\gamma = 1.45$ to obtain radii s_k^{gt} . This yields a ground-truth semantic model

$$\mathcal{B}^{\text{gt}} = \left\{ \left(X_k^{\text{gt}}, s_k^{\text{gt}}, q_k^{\text{gt}} \right) \right\}_{k=1}^{K_{\text{gt}}},$$

which is directly comparable to INIT and OPT predictions.

5.2 Evaluation Metrics

We report three complementary evaluation measures.

- Runtime

For each view subset V_M , we measure the Stage-3 wall-clock runtime T_M from the INIT model to the refined OPT- M model. This quantifies the runtime–accuracy trade-off as M increases.

- Berry Counting Accuracy

Let $N_{b,m}^{\text{pred}}$ be the predicted berry count for bunch b under method $m \in \{\text{INIT, OPT} - 10, \text{OPT} - 20, \text{OPT} - 30\}$, and let N_b^{gt} be the ground-truth count.

The mean absolute error (MAE) for method m is

$$\text{MAE}_m = \frac{1}{B} \sum_{b=1}^B |N_{b,m}^{\text{pred}} - N_b^{\text{gt}}|. \quad (2)$$

We report MAE over all bunches and, separately, over dense bunches with $N_b^{\text{gt}} \geq 60$ to highlight behaviour under heavy occlusion.

- 3D Geometric Accuracy

To evaluate the quality of the 3D geometry, we compare predicted semantic ellipsoids with ground truth via OBB-IoU. For each method m , predicted ellipsoids are first matched to ground truth by Hungarian assignment using center distance. For each matched pair (E_p, E_g) , we compute the OBB-IoU

$$\text{IoU}(E_p, E_g) = \frac{\text{Vol}(E_p \cap E_g)}{\text{Vol}(E_p \cup E_g)}. \quad (3)$$

Given an IoU threshold $\tau \in \{0.10, 0.25, 0.45\}$, a match with $\text{IoU} \geq \tau$ is counted as a true positive (TP); unmatched predictions and ground-truth ellipsoids are treated as false positives (FP) and false negatives (FN), respectively. Precision, recall, and F1 for method m at threshold τ are then

$$\text{Precision}_{m,\tau} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall}_{m,\tau} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1}_{m,\tau} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}. \quad (4)$$

We report these metrics for INIT and all OPT- M configurations; low thresholds emphasize center localization, whereas higher thresholds require accurate alignment in position, scale, and (where applicable) orientation.

Table 1: Average Stage-3 refinement time (minutes) for different view subsets V_M (30 bunches).

Selected Views, V_M	V_{10}	V_{20}	V_{30}
Stage 3 Average Time (min)	1.8	3.8	5.1

Table 2: Mean absolute error (MAE) of berry count for INIT (Stage 2) and OPT (Stage 3 with view subset V_M).

Range	Count	INIT (MAE)	OPT-10 (MAE)	OPT-20 (MAE)	OPT-30 (MAE)
< 40	2	0.5	0.5	0.5	0.5
40 < 49	9	1.1	1.1	1.1	1.1
50 < 59	13	2.8	2.6	2.7	2.7
≥ 60	6	10.5	8.0	8.2	4.8
Overall	30	3.7	3.1	3.2	2.5

Table 3: Oriented bounding box Intersection of Union (OBB-IoU) detection metrics (mean precision/recall/F1 in %, 30 bunches). INIT (Stage-2 output); O10, O20, and O30 = OPT (Stage-3 output) with 10, 20 and 30 selected views, respectively.

IoU(τ)	INITR	INITP	INITF1	O10R	O10P	O10F1	O20R	O20P	O20F1	O30R	O30P	O30F1
0.10	34.7	36.6	35.6	77.1	80.4	78.7	86.0	89.8	87.8	87.3	89.9	88.5
0.25	10.1	10.4	10.3	63.3	65.8	64.5	72.6	75.8	74.1	74.0	76.0	74.9
0.45	0.3	0.3	0.3	6.2	6.3	6.3	10.4	10.7	10.5	9.8	10.0	9.9

5.3 Runtime and Computational Efficiency

Stage 3 runtime increases approximately linearly with the number of views used in differentiable optimization (Table 1). Using $M = 10$ views provide fast refinement (≈ 1.8 min/bunch), while $M = 30$ yields the highest accuracy at increased cost (≈ 5.1 min/bunch).

These timings are measured from the Stage-2 initialization (INIT) to the converged refined model (OPT- M); Stage-1 and Stage-2 costs are shared across all settings and are therefore excluded.

5.4 Counting Accuracy and Topology Behavior

Table 2 reports berry-count MAE for the Stage-2 output (INIT) and the Stage-3 refined models (OPT-10/20/30). For small and medium bunches, Stage 2 already achieves low counting error, and Stage 3 largely preserves topology. The main gains appear in dense bunches (≥ 60 berries), where refinement significantly reduces under-counting, particularly at $M = 30$ (dense MAE: $10.5 \rightarrow 4.8$).

Topology updates are conservative: across 30 sequences most bunches retain the same berry count, and when changes occur, they are dominated by berry additions rather than deletions, consistent with the design of the clone-prune mechanism.

5.5 3D Geometric Accuracy Oriented Bounding Box Intersection of Union (OBB-IoU)

We evaluate 3D geometric quality using OBB-IoU between predicted and ground-truth ellipsoids. For a given IoU threshold τ , we match predictions to ground truth by Hungarian assignment on centers and compute precision, recall, and F1 over matched pairs.

Table 3 shows results for INIT and OPT- M at $\tau \in \{0.10, 0.25, 0.45\}$. Differentiable refinement yields substantial gains at $\tau = 0.10$ and $\tau = 0.25$, indicating much tighter alignment of berry centres and

scales. Performance at $\tau = 0.45$ remains challenging due to occlusion and noise in the ellipsoid ground truth but still improves over the Stage-2 initialisation.

Overall, Stage 3 improves F1 by roughly +43–53 points at $\tau = 0.10$ and +54–65 points at $\tau = 0.25$, while keeping refinement time under 6 minutes per bunch for $M \leq 30$.

5.6 Qualitative Comparison

Figure 4 illustrates a representative dense bunch (ground-truth: 55 berries). The labelled point cloud is converted into a semantic ground-truth model by fitting Gaussian ellipsoids to each berry (54 berries after minor manual consolidation). The Stage-2 initialisation already recovers a plausible topology but exhibits noticeable misalignment in some centres and orientations.

After Stage-3 refinement, models using V_M with $M \in \{10, 20, 30\}$ all converge to the correct berry count (54) and progressively improve geometric alignment, reaching IoU scores of 80.7 ($M=10$) and 89.9 ($M=20/30$) with runtimes of about 1.4, 4.2, and 6.2 minutes, respectively. The refined ellipsoids also correct several erroneous ground-truth orientations, indicating that the differentiable optimisation improves both position and shape under multi-view supervision.

6 CONCLUSION

We presented a three-stage framework for instance-level 3D semantic reconstruction of grape berries from short handheld multi-view sequences captured in real vineyards. Stage 1 extracts multi-view 2D supervision (centres, shape/scale, masks, boxes, visibility) and initial camera poses from a lightweight SfM COLMAP run. Stage 2 builds a compact 3D berry set via geometry-aware matching and triangulation with conservative validation and 3D consolidation.

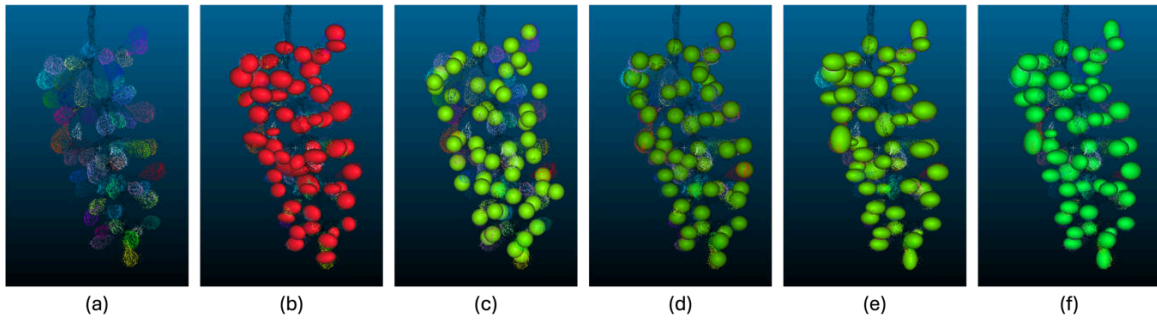


Figure 4: Qualitative comparison on a dense bunch (ground truth: 55 berries). (a) Labelled 3D point cloud. (b) Ground-truth semantic ellipsoids (55 berries). (c) Stage-2 initialization. (d–f) Stage-3 refined models using V_M with $M = 10, 20, 30$ views, respectively. Refinement preserves the correct berry count (54) while increasing IoU up to 89.9 with practical runtimes (~ 1.4 – 6.2 min).

Stage 3 refines 3D Gaussian berry primitives through differentiable rendering with multi-head 2D losses, combining a warm-up phase that corrects moderate pose drift with a conservative clone–prune mechanism for topology adaptation.

On 30 real-field bunches, refinement remains practical (≈ 1.8 – 5.1 min/bunch for $M \in \{10, 20, 30\}$), substantially reduces under-counting in dense bunches (dense MAE from 10.5 to 4.8 at $M = 30$), and significantly improves 3D alignment according to OBB-IoU detection metrics. Overall, the results indicate that combining geometry-aware initialisation with differentiable, instance-level refinement yields a compact and interpretable per-berry 3D representation suitable for unconstrained field imagery and downstream agronomic analysis.

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